

Build, govern and scale agentic AI | Think keynotes

What are AI agents?

AI agents

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By [Anna Gutowska](#)

AI agents defined

An [artificial intelligence \(AI\)](#) agent is performs tasks by designing [workflows](#)

AI agents can encompass a wide range of functions including decision-making, problem-solving, and performing actions.

AI agents solve complex tasks across enterprise automation, code generation and conversational language processing techniques of large language models respond to user inputs step-by-step and determine

For instance, Dynamiq built a [multi-agent legal](#) client using [IBM watsonx Orchestrate](#). The system uses a low-cost IBM Granite classifier first, escalating to a research agent. This pattern of intelligent routing reduced review time from 90 minutes to just 45 while being more effective.





What are AI Agents? (12:28 min)

How AI agents work

At the core of AI agents are [large language models](#), often referred to as LLM agents. Traditional LLMs generate their responses based on the data used to train them, with reasoning limitations. In contrast, agentic technologies obtain up-to-date information, optimize workflows, and achieve complex goals.

In this process, the autonomous agent learns from its agent's ability to store past interactions in [memory](#), creating a personalized experience and comprehensive understanding of the environment.

¹ This tool calling can be achieved without human intervention. [Components](#) define how agents operate:

Goal initialization and planning

Although AI agents are autonomous in their decision-making, they follow predefined rules defined by humans.² The agent behavior:

- The team of developers that design and train the agent.
- The team that deploys the agent and provides ongoing support.

- The user that provides the AI agent with specific goals and available tools to use.

Given the user's goals and the agent's available tools, the agent performs a task decomposition to improve performance.³ The agent breaks down tasks and subtasks to accomplish the complete goal.

For simple tasks, planning is not a necessary step. The agent can respond on its responses and improve them without planning.

Reasoning with available tools

AI agents base their actions on the information they have. If they lack the full knowledge required to tackle every situation, they turn to available tools such as external APIs or other agents.

When the missing information is gathered, the agent updates its plan and engages in [agentic reasoning](#). This process involves taking action and making self-corrections, which enables the agent to make better decisions.

To help illustrate this process, imagine a user agent with the goal of predicting which week in the next year is the best for their surfing trip in Greece.

Because the LLM model at the core of the agent cannot rely solely on its internal knowledge. The agent uses an external database containing daily weather data for the next years.

Despite acquiring this new information, the agent still needs to check the weather conditions for surfing and so, the next year the agent communicates with an external agent that provides weather data. By doing so, the agent learns that high tides and low winds are the best surfing conditions.

The agent can now combine the information it has gathered with its internal patterns. It can predict which week next year will have the best weather and a low chance of rain. These findings show that the sharing of information between tools is what sets AI agents apart from traditional AI models.³



Learning and reflection

AI agents use feedback mechanisms, such as (HITL) to improve the accuracy of their responses. For example, to highlight this process. After the agent completes a task, it reflects on the learned information along with the user's feedback to user preferences for future goals.

If other agents were used to reach the goal, the feedback can be especially useful in minimizing errors or in a new direction. However, users can also provide feedback on the agent's internal reasoning to better align the results with their goals.

Feedback mechanisms improve the AI agent's performance, which is referred to as iterative refinement.³ To avoid redundancy, agents also store data about solutions to previous objectives.

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Agentic versus nonagentic

[AI chatbots](#) use [conversational AI](#) techniques to understand user questions and automate responses. Nonagentic modality whereas agentic is a technological framework.

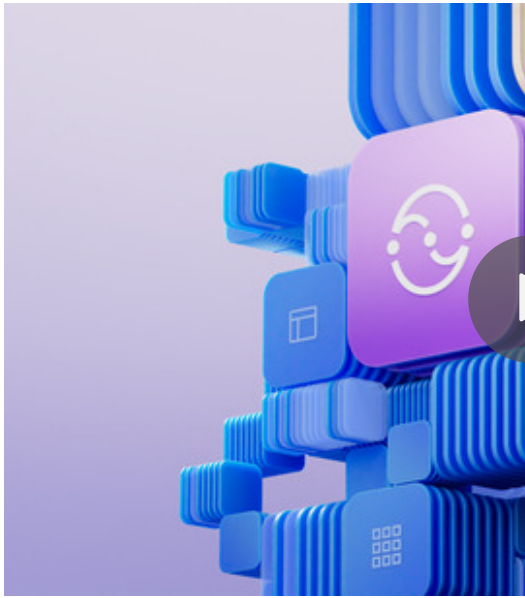
Nonagentic AI chatbots are ones without available context. They reach only short-term goals and cannot plan ahead. They require continuous user input to respond.

They can produce responses to common prompts but perform poorly on questions outside their scope. These chatbots do not hold memory, they cannot learn, and their responses are unsatisfactory.

In contrast, agentic AI chatbots learn to adapt to user needs, provide more personalized experience and comprehend complex tasks by creating subtasks without human intervention. These plans can also be self-corrected and updated. Unlike nonagentic ones, they assess their tools and use them to fill information gaps.

AI agents





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Reasoning paradigm

There is not one standard architecture for building solving multistep problems.

ReAct (reasoning and acting)

With the [ReAct paradigm](#), we can instruct agents to take actions and with each tool response to decide what to do next. Think-Act-Observe, are used to solve problem responses.

Through the prompt structure, agents can be instructed to think each “thought.”⁴ The agent’s verbal reasoning is formulated. In this framework, agents continue reasoning. This approach can be interpreted as



ReWOO (reasoning withc

The ReWOO method, unlike ReAct, eliminates planning. Instead, agents plan upfront. Redun which tools to use upon receiving the initial pr desirable from a human-centered perspective before it is executed.

The ReWOO workflow is made up of three mor anticipates its next steps given a user's promp outputs produced by calling these tools. Lastly outputs to formulate a response. This planning computational complexity and the repercussio

Types of AI agents

AI agents can be developed to have varying le preferred for straightforward goals to limit unr of simplest to most advanced, there are 5 mai

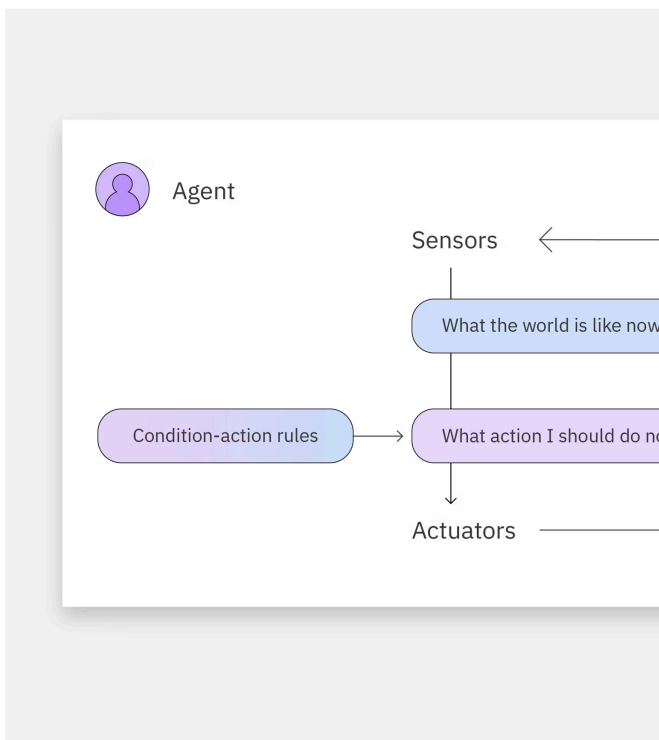
1. Simple reflex agents

Simple reflex agents are the simplest agent fo agent does not hold any memory, nor does it i information. These agents function on a set of means that the agent is preprogrammed to pe conditions being met.

If the agent encounters a situation that it is nc appropriately. The agents are effective in envi access to all necessary information.⁶

Example: If it is 8 PM, then the heating is acti on the heating system at a set time every nigh





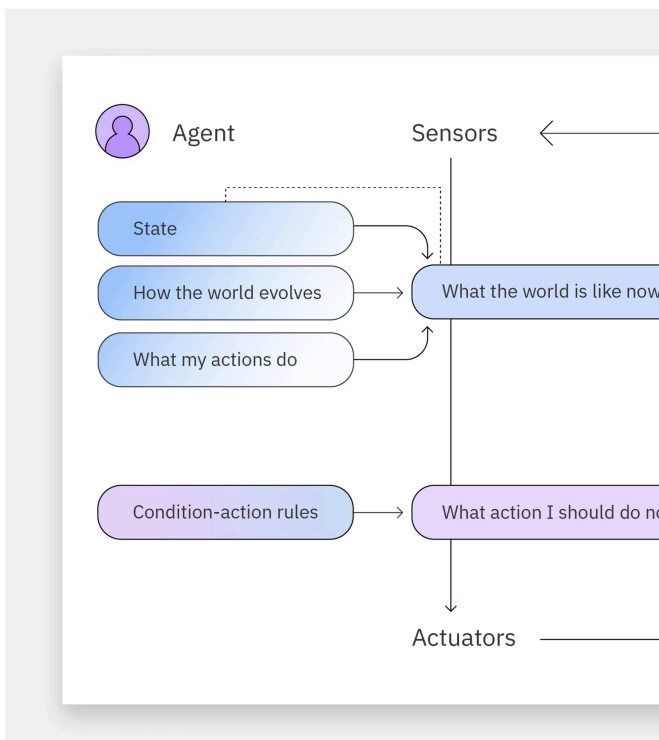
Simple reflex agent diagram

2. Model-based reflex ag

Model-based reflex agents use both their current internal model of the world. As the agent's current state is updated. The agent's actions depend on its current state.

These agents, unlike simple reflex agents, can operate in environments that are partially observable. Their actions are limited by their set of rules.⁶

Example: A robot vacuum cleaner. As it cleans, it moves around furniture and adjusts around them. The robot's actions are limited by its set of rules to not get stuck in a loop of re-

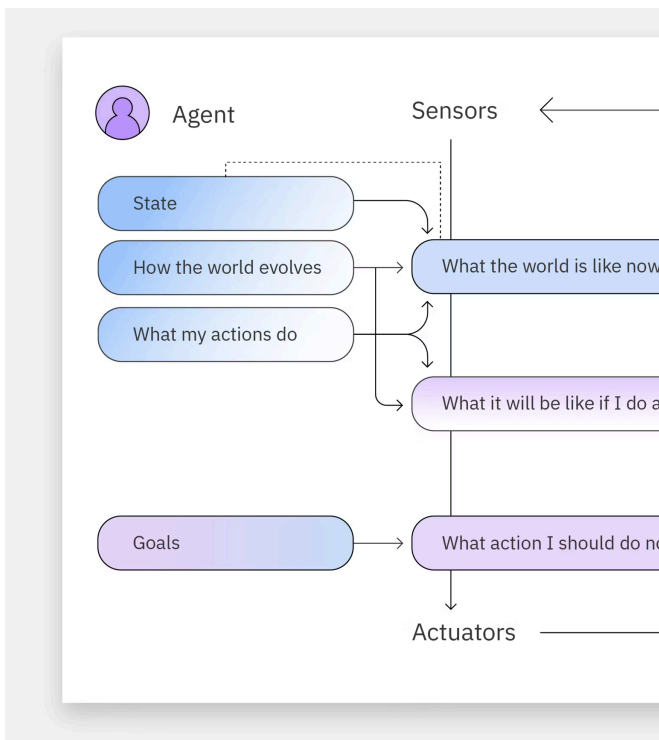


Model-based reflex agent diagram

3. Goal-based agents

Goal-based agents have an internal model of the world. These agents search for action sequences that achieve their goals before acting on them. This search and planning is more complex compared to simple and model-based reflex agents.

Example: A navigation system that recommends routes. The model considers various routes that reach your destination. In this example, the agent's condition-action rules are based on the agent's current location. The agent recommends that one instead.



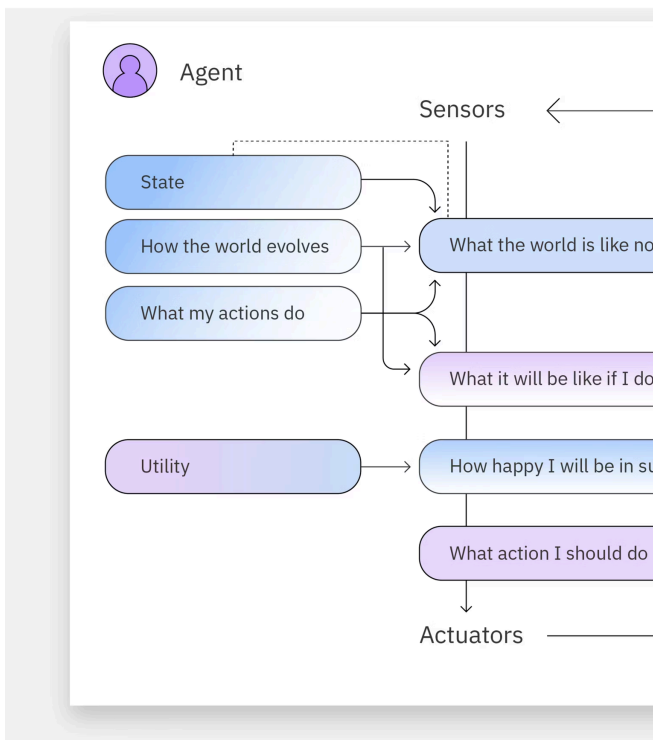
Goal-based agent diagram

4. Utility-based agents

Utility-based agents select the sequence of actions that maximizes the expected utility or reward. Utility is calculated through a value, a metric measuring the usefulness of an action in each scenario based on a set of fixed criteria.

The criteria can include factors such as progress towards the goal, computational complexity. The agent then selects the action with the highest utility. Hence, these agents are useful in cases where multiple actions can achieve a goal and an optimal one must be selected.⁷

Example: A navigation system that recommends the shortest path. The agent optimizes fuel efficiency and minimizes the time taken. The agent measures utility through this set of criteria.



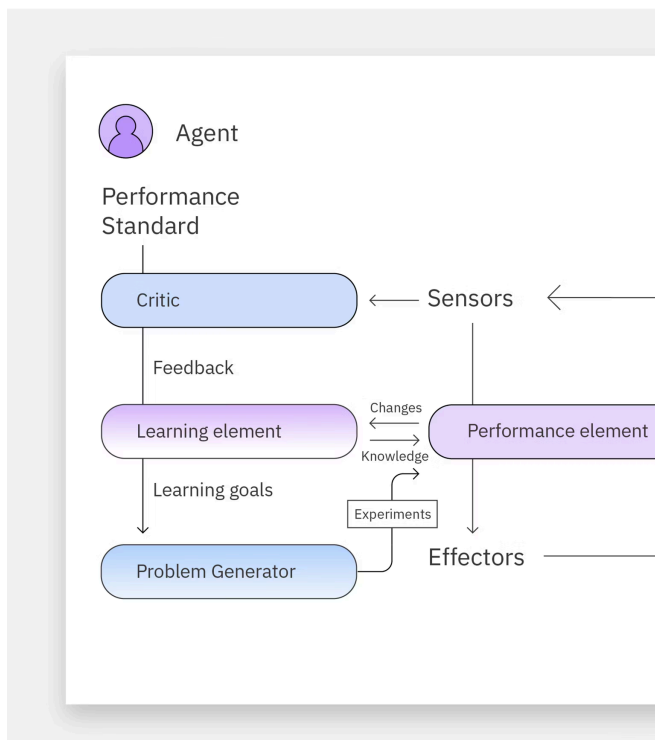
Utility-based agent diagram

5. Learning agents

Learning agents hold the same capabilities as ability to learn. New experiences are added to autonomously. This learning enhances the age environments. Learning agents might be utility composed of four main elements:⁷

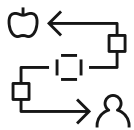
- **Learning:** This process improves the agent environment through its precepts and sens
- **Critic:** This component provides feedback responses meets the performance standar
- **Performance:** This element is responsible
- **Problem generator:** This module creates \

Example: Personalized recommendations on activity and preferences in their memory. This products and services to the user. The cycle re made. The user's activity is continuously store agent improves its accuracy over time.



Learning agent diagram

Use cases of AI agents



Customer experience

AI agents can be integrated into the customer experience by serving mental health support, simulating tasks.⁸ There are many no-code tools making the process of creating the



Healthcare

AI agents can be used for various **Multi-agent systems** can be used. From treatment planning for patients to managing drug processes, these medical professionals for more use



Emergency response

If there is a natural disaster, AI **learning** algorithms to retrieve the sites that need rescue. The location

assist rescue services in saving r
agents can greatly benefit huma
and life-saving situations.¹⁰



Finance and supply chain

Agents can be designed to analy
future market trends and optimi
customizability of autonomous A
personalized outputs to our uniq
data, it is important to enforce se



Reimagining productivity with AI agents and assis

Benefits of AI agents

Task automation

With the ongoing advancements in [generative](#)
interest in workflow optimization through AI, c
tools that can automate complex tasks that w
This shift converts to goals being reached inex
these advancements mean human agents do i
assistant for creating and navigating its tasks.

Greater performance

Multi-agent frameworks tend to outperform single-agent frameworks. As more plans of action are available to an agent, the more effectively it can perform.

An AI agent incorporating knowledge and feedback from related areas can be useful for information systems. Multi-agent systems and the ability to fill information gaps are becoming a powerful tool and a meaningful advancement in AI.

Quality of responses

AI agents provide responses that are more contextually relevant to the user than traditional AI models. This adaptability to user needs and higher-quality responses typically yield a better user experience. As described, this capability is made possible through multi-agent systems, through tools and updating their memory, and are not preprogrammed.¹²

Risks and limitations

Multi-agent dependence

Certain complex tasks require the knowledge and coordination of multiple agents. Multi-agent frameworks have a risk of malfunction if one agent fails. Some [foundation models](#) might experience shared system-wide failure of all involved agents or dependencies. This highlights the importance of [data governance](#), robust training and testing processes.

Infinite feedback loops

The convenience of the hands-off reasoning framework comes with its risks. Agents that are unable to learn from their findings, might find themselves repeated in feedback loops. To avoid these redundancies, multi-agent systems might be used.¹³



Computational complexity

Building AI agents from scratch is both time- and resource-intensive. The resources required for training and deployment can be significant. In addition, depending on the complexity of the tasks, the agents may require a large amount of data to learn from effectively.¹²

Data privacy

If mismanaged, the integration of AI agents with enterprise management systems can raise some serious concerns, such as data privacy and security. AI agents leading the software development process or determining pricing for clients—without any human oversight—such scenarios might be detrimental due to the unpredictable behavior of [agentic AI](#).

Therefore, it is essential for AI providers such as IBM to be proactive. They must implement extensive security measures to ensure that employee and customer data are securely stored and processed. This is key to minimizing risk and maintaining trust in the system.

Best practices

Activity logs

To address the concerns of multi-agent dependencies, it is recommended to have access to a log of agent actions.¹⁴ The actions recorded in the logs describe the external agents used to reach the final decision. This transparency into the iterative decision-making process, particularly when combined with other best practices, helps build trust.

Interruption

Preventing autonomous AI agents from running indefinitely is recommended. Particularly, in cases of unintended actions, such as unauthorized access to certain tools, or malfunctioning due to a lack of clear goals, the goal is to implement interruptibility.



Maintaining control of this decision involves allowing an AI agent to interrupt a sequence of actions or the entire operation. An AI agent requires some thoughtfulness as to what is good. For instance, it might be safer to allow a system to handle a threatening emergency than to completely shut it down.

Unique agent identifiers

To mitigate the risk of agentic systems being used for malicious purposes, unique identifiers can be implemented. If these identifiers are linked to external systems, tracing the origin of the agent's actions becomes easier.

This approach adds an essential layer of accountability by identifying responsible parties when an agent causes malfunctions. This kind of safeguard would foster a safer operational environment.

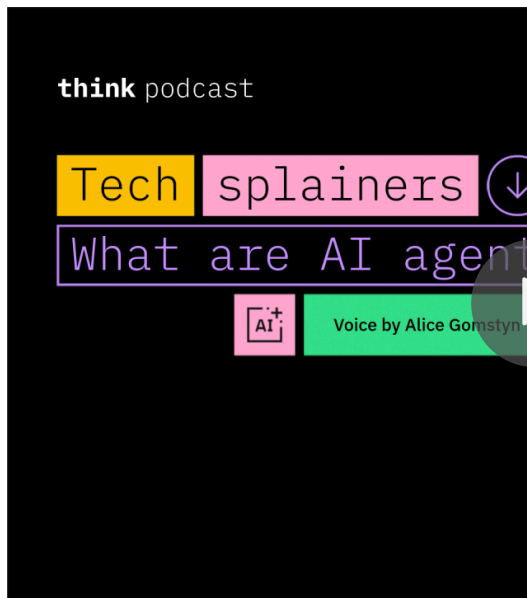
Human supervision

To assist in the learning process for AI agents, human oversight is crucial. In a supervised environment, it can be helpful to provide some initial guidance, after which the AI agent can compare its performance against human preferences and make adjustments. This form of feedback is helpful for refining the agent's preferences.⁵

Apart from this safeguard, it is best practice to have human oversight when an agent takes highly impactful actions. For instance, AI agents used in emails to financial trading should require human monitoring. Human monitoring is recommended for such high-risk applications.

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Advocate
IBM



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Topic

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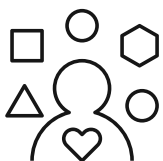
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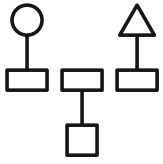
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